

INTRODUCTION TO DATA SCIENCE AND PROGRAMMING

BRENDAN MCCANN

A REAL-LIFE EXAMPLE

```
password = input("enter password: ")

while len(password) <= 4:
    print("invalid password!")
    password = input("enter password: ")

print("password created successfully!")
```

- **What does this program do?**

A REAL-LIFE EXAMPLE

```
password = input("enter password: ")

while len(password) <= 4 or "!" not in password:
    print("invalid password!")
    password = input("enter password: ")

print("password created successfully!")
```

- What if I wanted to make the password a bit more complicated?

A REAL-LIFE EXAMPLE

```
password = input("enter password: ")

while len(password) <= 4 or "!" not in password or password == "12345":
    print("invalid password!")
    password = input("enter password: ")

print("password created successfully!")
```

- What if I wanted to make the password even more complicated?

FUNCTIONS

A REAL-LIFE EXAMPLE

```
password = input("enter password: ")

while isPasswordInvalid(password):
    print("invalid password!")
    password = input("enter password: ")

print("password created successfully!")
```

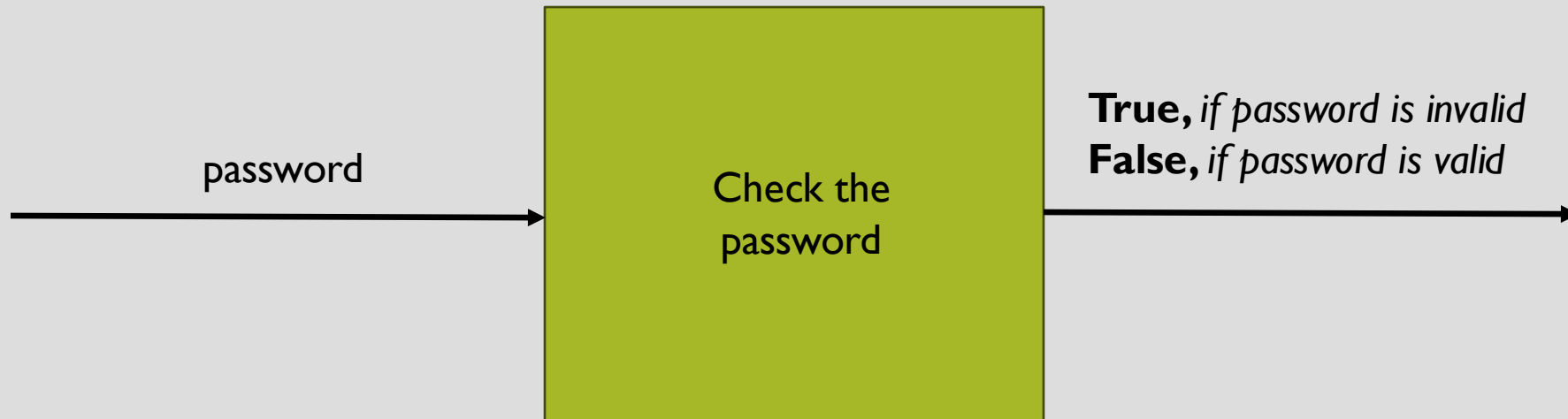
A REAL-LIFE EXAMPLE

```
def isPasswordInvalid(password):  
    if len(password) <= 4:  
        return True  
    elif "!" not in password:  
        return True  
    elif password == "12345":  
        return True  
    else:  
        return False
```

WHAT IS A **FUNCTION**?



ISPASSWORDINVALID() FUNCTION



Name of function

Input

"def" - short for "define"
Tells python this is a
function

Outputs

```
def isPasswordInvalid(password):  
    if len(password) <= 4:  
        return True  
    elif "!" not in password:  
        return True  
    elif password == "12345":  
        return True  
    else:  
        return False
```

A REAL-LIFE EXAMPLE

```
password = input("enter password: ")

while isPasswordInvalid(password):
    print("invalid password!")
    password = input("enter password: ")

print("password created successfully!")
```

```
def isPasswordInvalid(password):
    if len(password) <= 4:
        return True
    elif "!" not in password:
        return True
    elif password == "12345":
        return True
    else:
        return False
```